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Diffusion Policy with Behavioral Advantage Correction for Offline Reinforcement Learning

BAC-PE Uses the Behavior Policy’s Q-Function to Simultaneously Tame Overestimation and Conservatism in Offline RL

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Offline reinforcement learning is caught between two failure modes: over-conservative penalization of in-distribution actions and overestimation of out-of-distribution actions. Behavioral Advantage Corrected Policy Evaluation (BAC-PE) resolves this dilemma by adding the behavior policy’s advantage function as a correction term to the learned Q-function, with convergence guarantees and strong empirical results across D4RL, Antmaze, and robotic manipulation benchmarks.

AT A GLANCE

Problem: Distribution shift in offline RL causes Q-functions to either overestimate out-of-distribution actions or over-penalize in-distribution ones.

Why it matters: Safe offline RL from logged data is essential for robotics and healthcare, where online interaction is expensive or dangerous.

Method: Correct the learned Q-function with the behavior policy’s advantage; represent both policies as diffusion models for multimodal action distributions.

Result: Outperforms CQL, TD3+BC, and Diffusion-QL on D4RL locomotion and Antmaze; strong results on robotic manipulation.

Evidence: Theoretical convergence bound on Q-estimation error; evaluated on D4RL, Antmaze, and Robosuite benchmarks.

The Big Picture

Offline reinforcement learning promises to extract useful policies from logs of past behavior—a critical capability for domains such as healthcare and autonomous driving where online interaction is too costly or dangerous. The fundamental challenge is that the learned policy will inevitably visit state-action pairs not well represented in the fixed dataset. Standard Q-learning then propagates erroneous value estimates for these out-of-distribution (OOD)

actions, causing the policy to exploit them catastrophically.

The field has responded with two families of corrections. *Conservative* methods (CQL, IQL) penalize OOD actions with explicit regularization, but the penalty is necessarily blunt: it can suppress well-covered in-distribution actions alongside genuinely dangerous OOD ones. *Implicit constraint* methods (TD3+BC, BEAR) restrict the policy to stay close to the behavior distribution, but tight constraints sacrifice the ability to improve over a mediocre behavior policy. BAC-PE takes a fundamentally different approach: rather than penalizing OOD actions by their distance from the behavior distribution, it *corrects* Q-values using information the behavior policy actually provides.

Why Existing Approaches Fall Short

The behavior policy μ generates all data in the offline dataset. Its Q-function Q^μ is estimable from the data without distribution shift, because every action in the dataset was taken by μ on the state distribution that μ itself induced. This is the key observation that existing methods leave on the table.

Conservative Q-Learning (CQL) adds a penalty $-\mathbb{E}_{a \sim \pi}[Q]$ to push Q-values down for policy actions and up for data actions. But it does so uniformly across all policy actions, regardless of whether they are genuinely OOD or merely low-density actions that μ took rarely but which are still valid.

TD3+BC adds a behavioral cloning regularizer $\|a - a_\mu\|^2$ to the policy gradient, preventing large deviations from the logged actions but also preventing the policy from taking shortcuts the behavior policy missed.

Neither method uses Q^μ directly. BAC-PE does.

Core Mechanism

Behavioral Advantage. The behavioral advantage of action a in state s is:

$$A^\mu(s, a) = Q^\mu(s, a) - V^\mu(s),$$

where $V^\mu(s) = \mathbb{E}_{a \sim \mu}[Q^\mu(s, a)]$. A positive behavioral advantage means a is *better than average* under μ ; a negative behavioral advantage means a is OOD or poor.

BAC-PE Correction. The corrected Q-function for the learned policy π is:

$$\hat{Q}^\pi(s, a) = Q^\pi(s, a) + \alpha A^\mu(s, a).$$

The intuition:

- If a was frequently taken and good under μ (large $A^\mu > 0$), $\hat{Q}^\pi$ is boosted, offsetting excessive conservatism.

- If a was rare or poor under μ (small or negative A^μ), $\hat{Q}^\pi$ is suppressed, offsetting overestimation.

No uniform penalty is applied; the correction is adaptive, action-by-action.

Diffusion Policy. Both μ and π are represented as *diffusion models* over the action space:

$$\pi_\theta(a|s) \propto \int p_{\theta,T}(a_T) \prod_{t=1}^T p_{\theta,t}(a_{t-1}|a_t, s) da_{1:T},$$

where $a_T \sim \mathcal{N}(0, I)$ and $p_{\theta,t}$ is a learned denoising step. Diffusion policies capture multimodal action distributions (e.g. multiple valid grasps for a manipulation task) that Gaussian or deterministic policies cannot.

Technical Formulation

Let $\mathcal{D} = \{(s_i, a_i, r_i, s'_i)\}$ be the offline dataset collected under μ . BAC-PE estimates Q^μ and V^μ via standard TD with dataset transitions:

$$Q_\phi^\mu \leftarrow \arg \min_{\phi} \mathbb{E}_{\mathcal{D}} \left[\left(Q_\phi^\mu(s, a) - r - \gamma V_\phi^\mu(s') \right)^2 \right],$$

then learns Q^π with the corrected target:

$$\hat{Q}_\psi^\pi \leftarrow \arg \min_{\psi} \mathbb{E}_{\mathcal{D}} \left[\left(\hat{Q}_\psi^\pi(s, a) - r - \gamma \hat{Q}_\psi^\pi(s', a') \right)^2 \right],$$

where $a'_{s'} \sim \pi_\theta(\cdot|s')$.

The policy is optimized to maximize $\hat{Q}^\pi$ subject to a KL constraint on deviation from μ :

$$\mathcal{L}(\theta) = -\mathbb{E}_{s,a \sim \pi_\theta} \left[\hat{Q}^\pi(s, a) \right] + \lambda D_{\text{KL}}(\pi_\theta(\cdot|s) \parallel \mu(\cdot|s)).$$

Learning or Inference Procedure

Stage 1: Behavior policy fitting. A diffusion model is fit to the offline dataset actions conditioned on states: this is standard behavioral cloning with a diffusion denoiser.

Stage 2: Q-function estimation. Q_ϕ^μ and V_ϕ^μ are trained via TD using the offline dataset.

Stage 3: Policy optimization. The diffusion policy π_θ is optimized alternately updating the denoising network (to maximize $\hat{Q}^\pi$) and the Q-function Q_ψ^π (to provide targets for the next policy update).

Inference. At test time, the diffusion policy generates actions by running T denoising steps from Gaussian noise conditioned on the current state.

What the Guarantee Says

Theorem (informal). Under standard concentrability and function approximation assumptions, the BAC-PE Q-function Q_k^π at iteration k satisfies:

$$\|Q_k^\pi - Q^{\pi^*}\|_\infty \leq \varepsilon_k + C \delta_\mu,$$

where $\varepsilon_k \rightarrow 0$ as $k \rightarrow \infty$, C is a problem constant, and $\delta_\mu = \sup_{s,a} (\pi(a|s)/\mu(a|s))$ is the concentrability coefficient. Actions well-covered by μ (δ_μ small) converge to their true Q-values; error grows gracefully as actions move OOD.

Experimental Findings

- **D4RL locomotion** (HalfCheetah, Hopper, Walker2D, Ant): BAC-PE improves over Diffusion-QL by 3–7 normalized score points on medium-expert datasets, the most challenging offline distribution.
- **Antmaze** (medium-play, large-diverse): 8–12 success rate improvement over CQL, which is excessively conservative on long-horizon tasks.
- **Robosuite manipulation** (Lift, PickPlace): 15% improvement in task completion over TD3+BC, where multimodal action distributions are critical.

Ablations and Interpretation

Advantage coefficient α . Setting $\alpha = 0$ (no correction) gives standard diffusion Q-learning; increasing α improves performance up to $\alpha \approx 1$ and then degrades, consistent with the theoretical bound: too large α overweights behavior-policy information at the expense of the true Q-values.

Diffusion vs. Gaussian policy. Replacing the diffusion policy with a Gaussian policy degrades performance by 5–8 points on manipulation tasks where action distributions are strongly multimodal, confirming that representation quality matters alongside the BAC-PE correction.

Q^μ estimation quality. Using a poorly estimated Q^μ (trained on a small data subset) degrades performance, confirming that BAC-PE’s advantage is contingent on reliable behavior-policy value estimation.

Reference

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